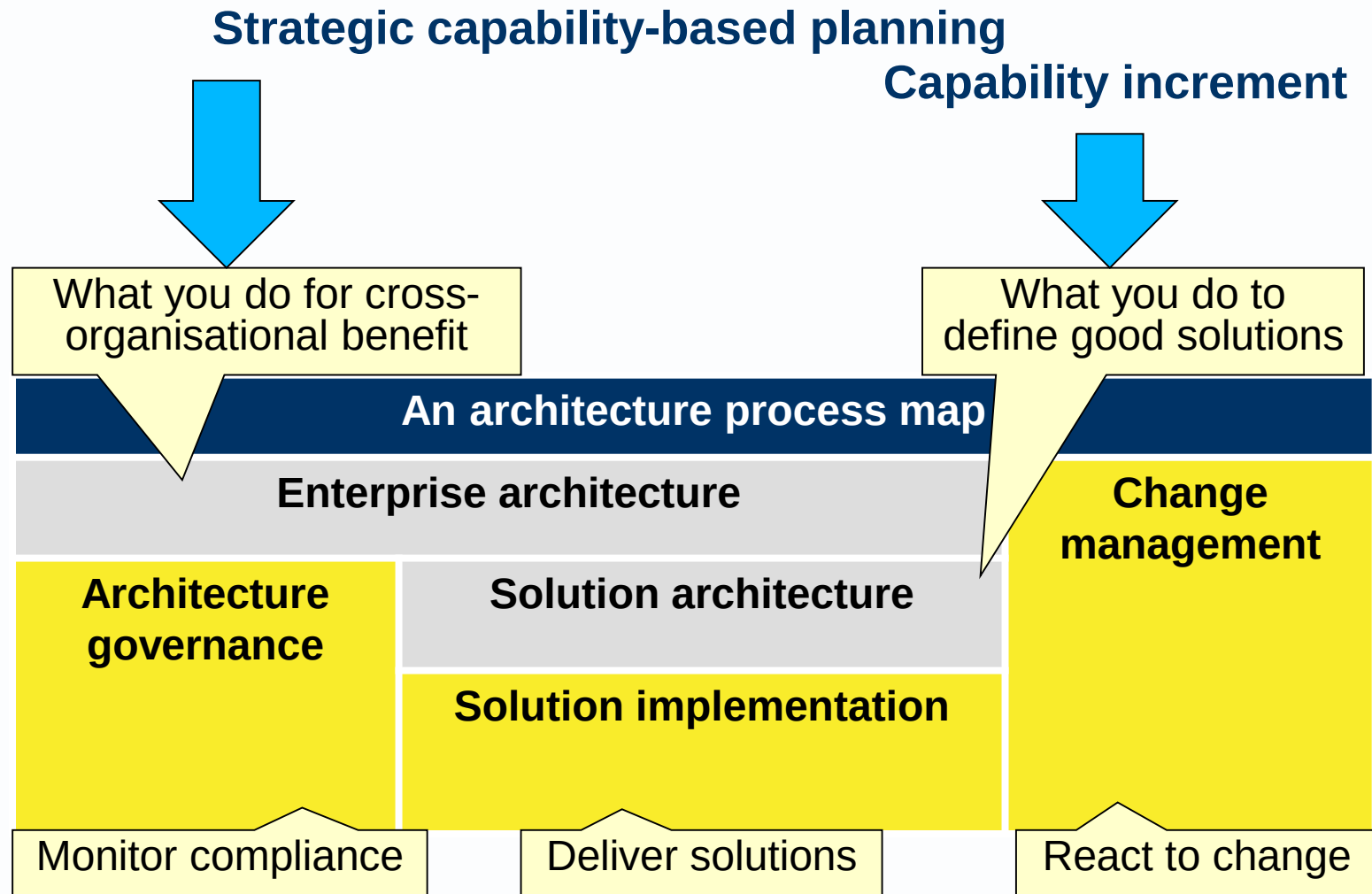


Avancier Reference Model

Architecture Management (ESA 11)

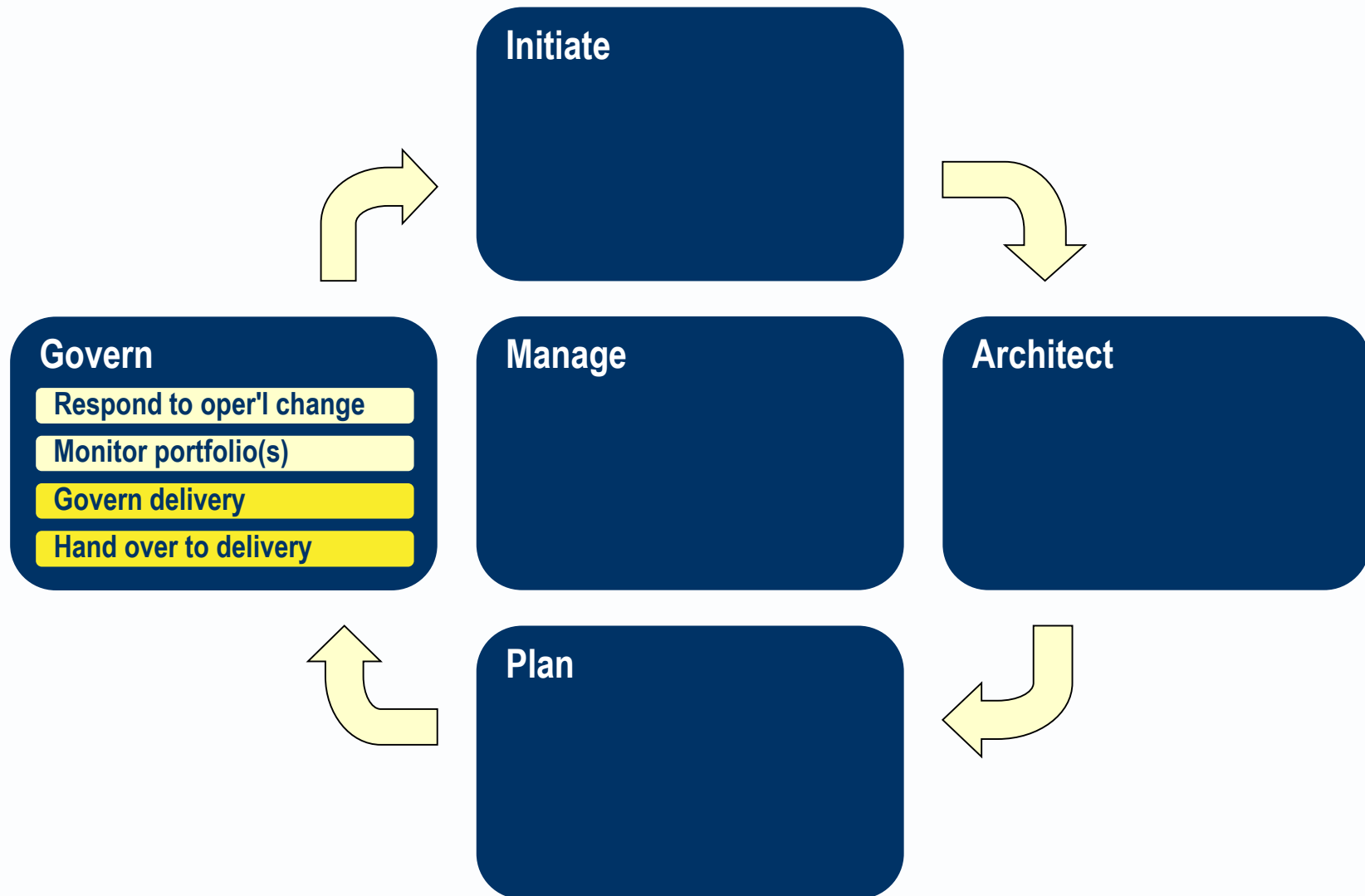
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The organisation and processes that are needed to **govern** and **implement** an architecture description, both in development and in operation, including the management of **changes**.

	People	Processes	Products
General Management	Board Manager Team	Set aims and directives Plan Start/initiate Monitor Control Stop/close	Aims and directives Plans Review criteria Progress reports Conclusions Process definitions

Architecture Implementation - in the AM process



- [A work process] that realise an architecture through system development and deployment.
- This requires programme and project management roles and processes.

Software Development Life Cycle (SDLC)

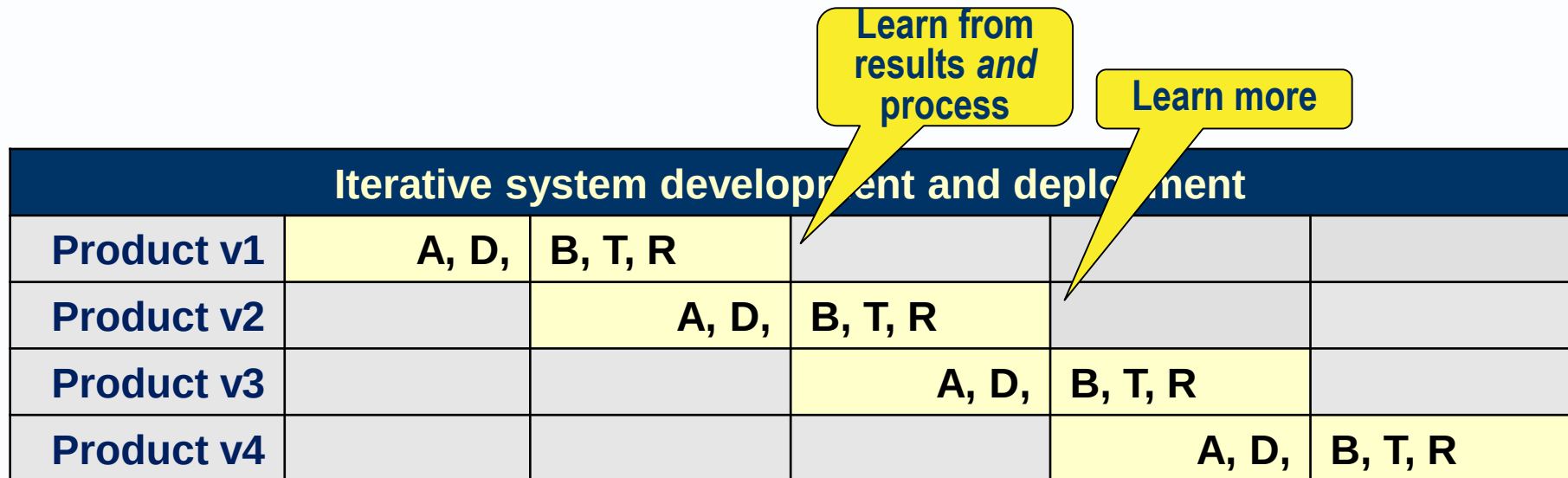
- [A work process] centered on analysis, design software engineering, testing and roll out.
- There are agile, iterative and waterfall variants.

	People	Processes	Products
SDLC	Steering group Project manager Project team	Plan projects Initiate projects SDLC Phase/milestone reviews Close projects	Project plans Project initiation documents Review checklists Review reports Project closure reports Process definitions

- [A technique] a development process that places analysis, design, build, test and roll out in sequence.
- Engineers proceed from one kind of work to the next without significant iteration or parallelism between stages.

Waterfall system development and deployment					
Analyze	A				
Design		D			
Build			B		
Test				T	
Roll out					R

- [A technique] a development process that proceeds by increments, meaning that a working subset of the full solution is delivered as early as possible.
- It is a foundation of the Unified Method and known as Incremental Development in DSDM.
- It is an essential feature of agile methods, and may be used in non-agile projects also



Iterative system development and deployment					
Product v1	A, D,	B, T, R			
Product v2		A, D,	B, T, R		
Product v3			A, D,	B, T, R	
Product v4				A, D,	B, T, R

- [A technique] a solution development process that is not only iterative, but also flexible about the requirements, the solution and the process being followed.
 - It favours
 - negotiation over planning, and flexibility about requirements.
 - early testing for usability and performance; user involvement and feedback is a prerequisite.
 - short-cycle iterative development; it looks for the minimum change that adds value to a system, and strives to deliver that change in the next sprint/release.
 - capitalising on the skills and knowledge of a small team
- *“I estimate that 75% of those organizations using Scrum will not succeed in getting the benefits that they hope for from it.”*
 - Ken Schwaber in an interview posted on Agile Collab

What does agile mean?

- Fail faster is good!
- Delivery early, commit late.
- Accept flexible, prioritised and ever changing requirements.
- High-level documentation of specifications and models.
- Test-driven rather than model-driven.

Waterfall methods suggest model > code > test.

Agile methods suggest test > code > model.

- *[Illegible text]*
- *[Illegible text]*

Martin Fowler

BUT IN THE FIRST PLACE Does your project suit agile?

- Score your current or most recent project using the form below
- High scores make agile methods difficult (but not a bad thing)

What kind of project?		
Time/cost-driven	0, 1, 2, 3	Mandatory requirements-driven
Users available for Joint App Dvlpmnt	0, 1 , 2, 3	Users not available
Developers empowered	0, 1, 2, 3	Developers not empowered
What kind of system and work?		
Divisible into usable releases (soft target)	0, 1, 2, 3	Indivisible (hard target)
Simple UX and/or UI technology	0, 1, 2, 3	Complex UX and/or UI technologies
Output/enquiry/report dominated	0, 1, 2 , 3	Input/update dominated
Simple (CRUD) data processing rules	0, 1, 2, 3	Complex data processing rules
On-line	0, 1, 2, 3	Batch
Stand-alone	0, 1 , 2, 3	Highly integrated w other systems
Add up the scores for your agile potential quotient		

Agilists favour (points to 'Developers empowered')

Difficult projects (points to 'Mandatory requirements-driven' and 'Users not available')

Difficult projects (points to 'Developers not empowered')

Role of solution architect wrt agile software development

- Complete a solution outline in accord with EA
- Pick the platform technologies
persistent database structures
- Use agile development as far as possible
- Govern acceptability of changes during an agile project.

Can agile be effective with outsourcing?

12 Lessons Learned by Peter DeYoe www.it-insight-blog.com



- **Lesson 1:** [P ebar ibZ k l sbó m l c qib p q / e l r q d ovl r o l kpel d k ka l a pel d d j p t l d hkd a v) f c m p p f ib Q e f] greatly increased the communication flow and cohesiveness of the teams.
- **Lesson 2:** Create a robust repository and collaboration site that will be the site of record for all specifications, test cases and discussions. SharePoint [recorded] all communication, collaboration and critical Artefacts.
- **Lesson 3:** Do not use email as your *primary* method of communication for topics such as requirements clarification or design decisions. Ensure communication is conducted through the repository.
- **Lesson 4:** Implement web conferencing to create a sense of proximity. We used this daily to conduct stand ups, review wireframes and specifications, walk through requirements and conduct Sprint reviews.
- **Lesson 5:** Have one central point of entry for project status. Each team member records progress via a central Scrum management tool - we used ScrumWorks to create accurate product backlogs, Sprint backlogs and burn-downs on a daily basis.
- **Lesson 6:** Ensure your offshore team has their own development and test environments, bug reporting tools (Bugzilla in our case) and source code repository.
- **Lesson 7:** Shorten your Sprints. We shortened the Sprints from 4 to 2 weeks.
- **Lesson 8:** The Scrum Master - the key arbitrator - must be top notch - perhaps co-located with Product Owner.
- **Lesson 9:** If possible, your Scrum Master should speak the languages of onshore and offshore teams.
- **Lesson 10:** Q e b M d a r q l t k b o j r p q i b o v a b d k b t e q a l k b - j b k p d o b e r p b o p d o T b i i - defined acceptance criteria should be included in each user story.
- **Lesson 11:** A strong technical leader on the offshore team with all of the technical skills to be self sufficient.
- **Lesson 12:** The offshore team must be properly trained in Scrum and specifically in your particular implementation of Scrum.
- Additional thoughts on running an Agile project using offshore resources can be found in an article written by Martin Fowler <http://martinfowler.com/articles/agileOffshore.html>

- [A work process] that, once the architecture has been realised in the form of an operational system, hands over that system to the organisations that will operate it.

Transition into business operations

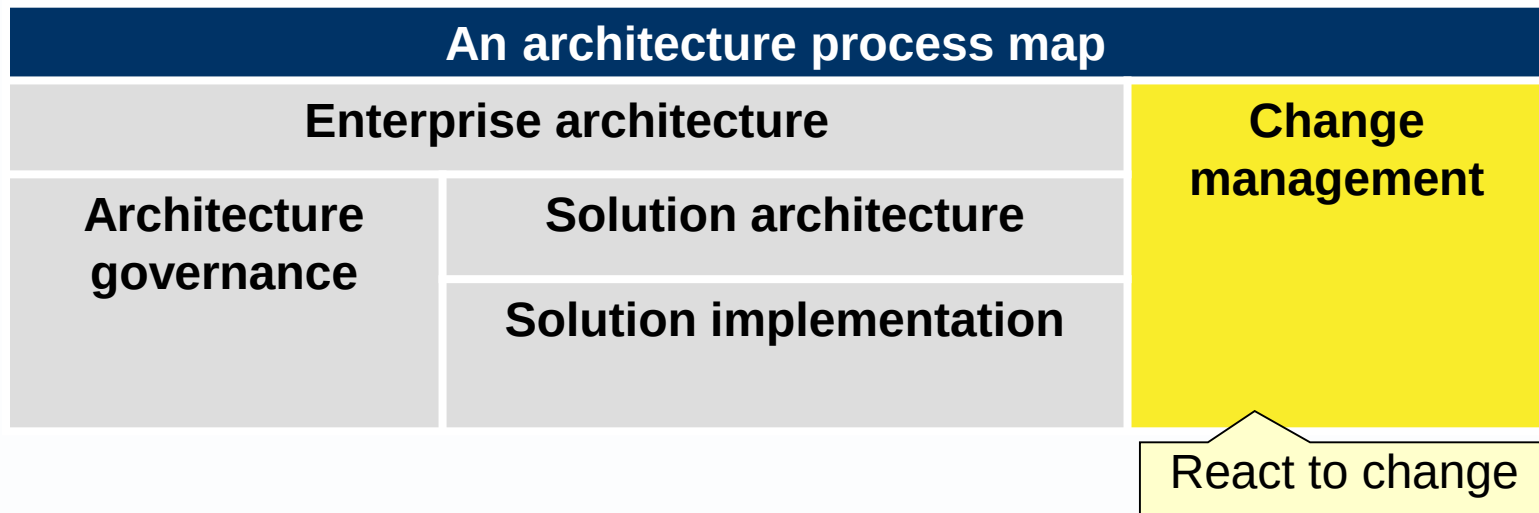
- [A work process] that hands over a completed solution to a business unit for business process operation and management.

Transition into IT operations

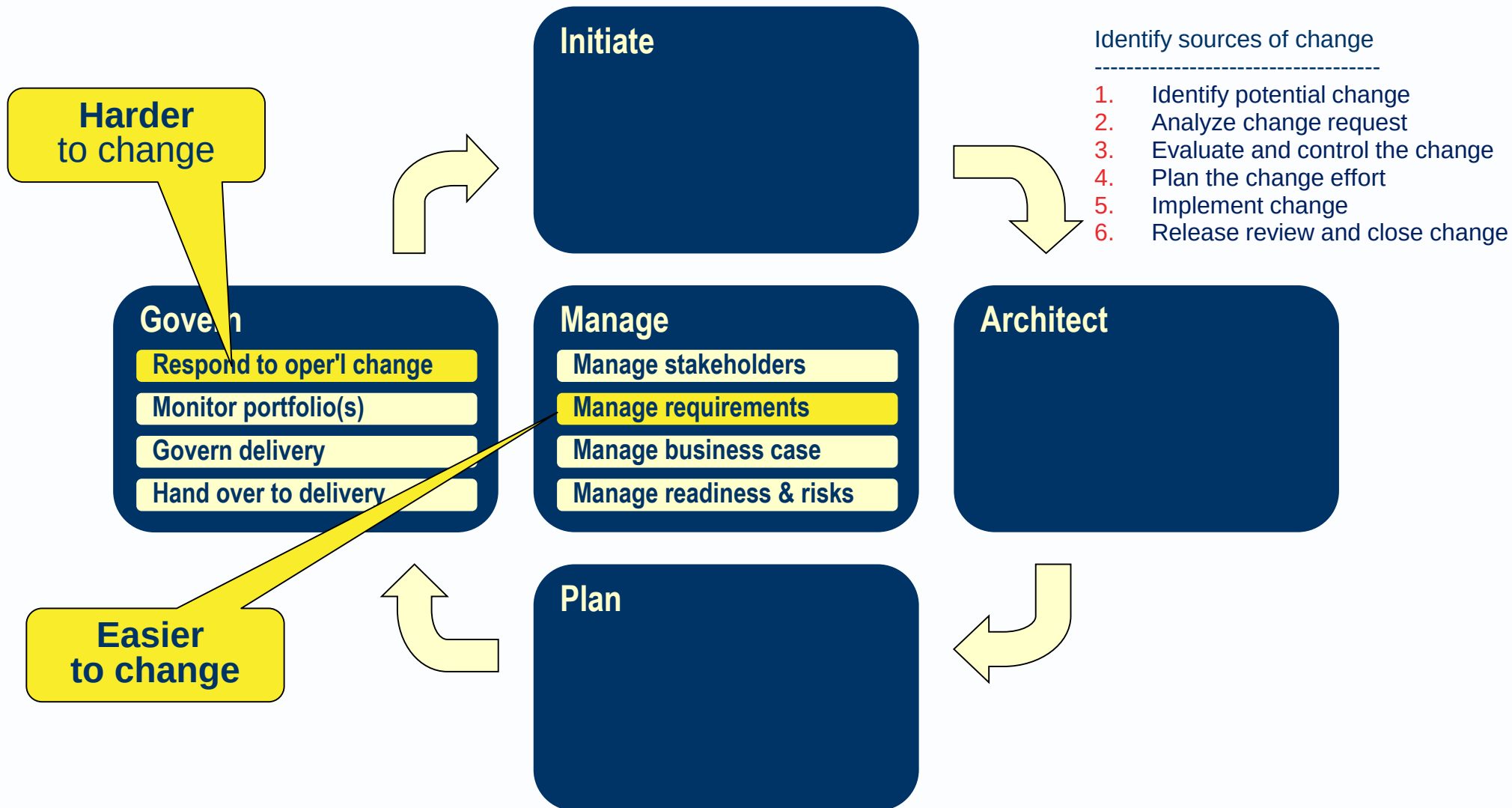
- [A work process] the hands over a production system to be run by managed operations or ITSM organisation.

Transition into maintenance

- [A work process] that hands over design and compile-time system to be maintained and perhaps enhanced by some kind of maintenance organisation.



Architecture change management - in the AM process



- [A technique] the roles and processes needed to both
- exercise **change control** to a baseline, and
- perform **configuration management**.

	People	Processes	Products
Change Management	CAB Change reviewers Review administrators	Change control, Monitor events Record change requests Analyse impacts of change Determine actions Process changes Configuration management.	Baseline configurations Configuration items Requests for change (RFC), Impact statements

■ **Change Control**

- [A technique] featuring the roles and processes needed within change management to
 - monitor potential sources of change
 - record change requests
 - perform impact analysis
 - determine, make and release changes

RFC: Request for Change

[An artifact] or form used to record details of a new requirement, a problem report or a change request to any configuration item.

Impact analysis

[A technique] to analyse the effects of a change, determine feasibility, update the business case and produce an impact analysis report.

- [A technique] the roles and processes needed within change management to establish a baseline configuration and apply changes to that baseline configuration.
- Involves work to:
 - Identify and document the characteristics of each item.
 - Define dependencies between items.
 - Control the introduction of new versions of items.
 - Report the status of configuration items and changes to them.

Baseline configuration

- [A passive structure] a specification or product that has been formally reviewed and agreed upon.
- The basis for further development.
- Can be changed only through formal change management.
- E.g. a contract, a requirements catalogue, architecture documentation, or a hardware configuration.

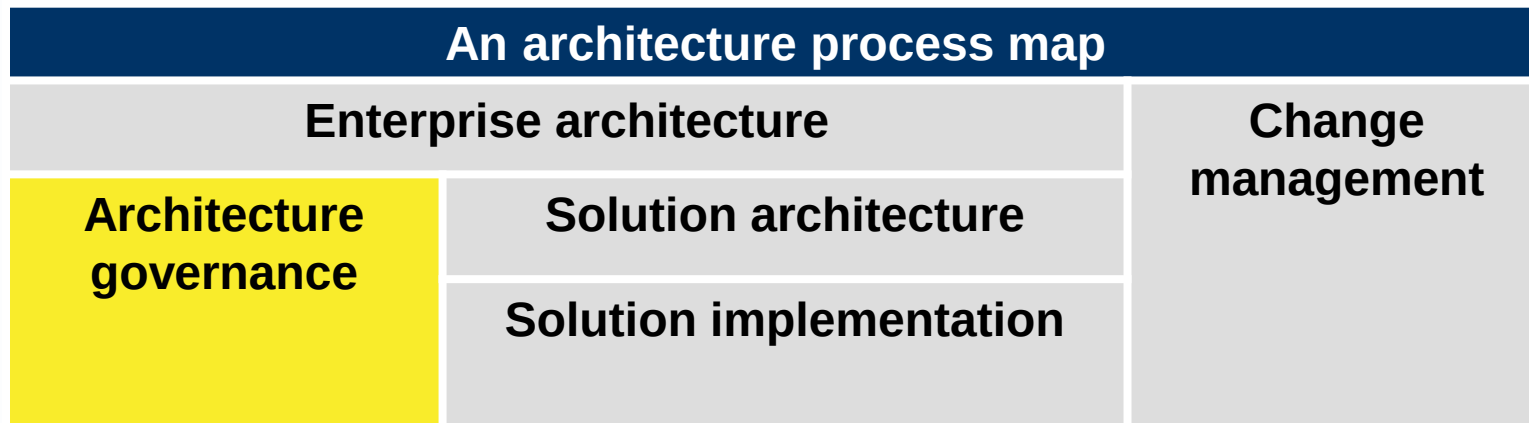
Configuration Item

- [A data object] an item in a baseline configuration.
- It could be a requirement, a source code component or a hardware device.
- It can be at any level of granularity.
- > I j m k b k q l c k f k o p o r q d r k a b o c e b l k o l i c l k d r o q l k management.
- A configuration item can range from an entire system (hardware, software, al r j b k q q l k & q p k d i b e o a t d l j m k b k q - F Q F

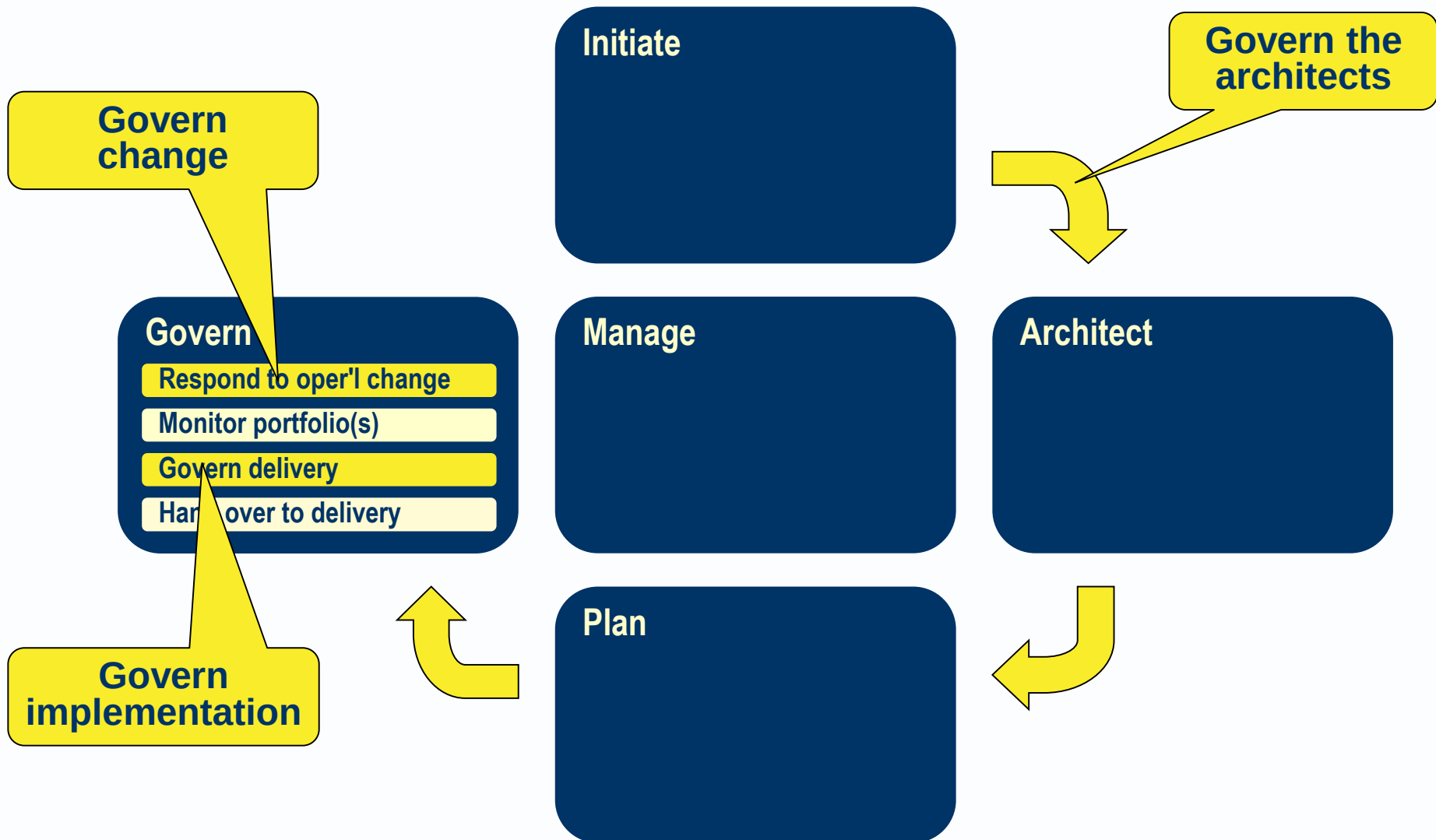
- [A work process] needed to manage changes to architectures, mostly stemming from changes to requirements or constraints, or operational systems.
- No different in principle from general change management (below).

- [A property] willing and able to speedily respond to change.
- Change management can turn into a bureaucracy that not only stifles change but also stifles progress
- Change management can be heavy or light
- The trick is to be agile without abandoning change control

- **Strategy & Architecture Board**
- *Architecture change control*
 - CIO
 - Business rep(s)
 - PMO rep(s)
 - Enterprise architect or lead solution architect
 - Domain specialists (business, data, apps, infrastructure, security)



Architecture governance in the AM process



What does a governor do?

- BBC?
- School?
- Building site?

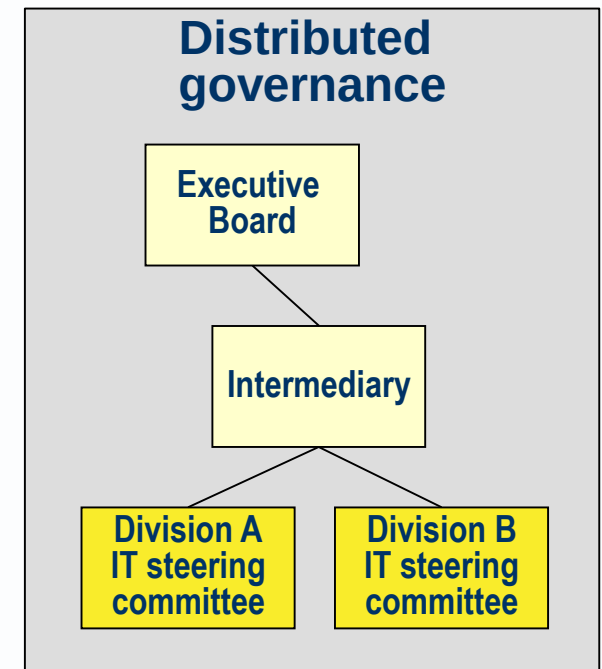
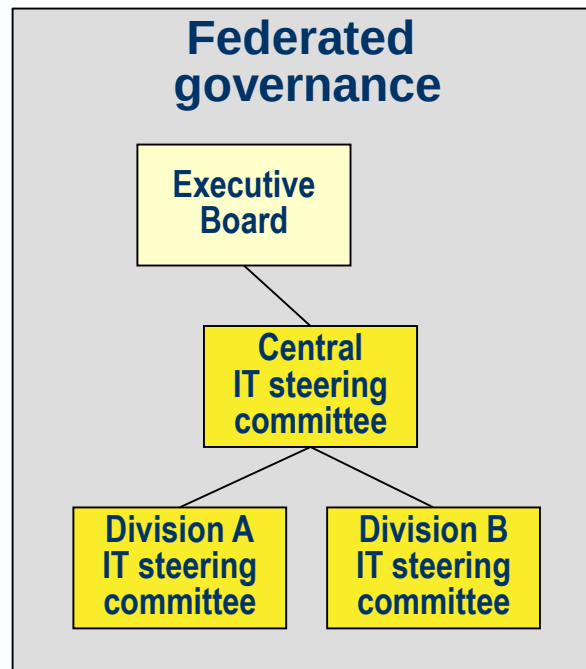
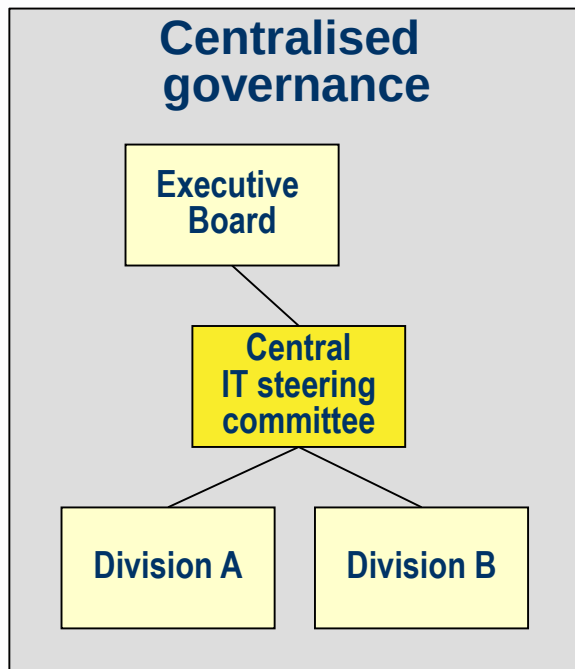
- Monitor compliance of operations against strategic principles, policies and goals

	People	Processes	Products
Governance	Board Governing architect Compliance reviewers Review administrators	Directive adoption & review Compliance review Dispensation Monitor and report Business alignment Environment management	Principles, policies, standards etc. Reference models Contracts Review checklists Review reports Dispensations Process definitions

- [A discipline] for monitoring and steering the management of an enterprise in accord with overarching drivers, goals/objectives and principles/policies.
- It may be subdivided into:
 - Corporate governance:** the responsibility of a corporate board or executive board. (SOX, Cadbury)
 - IT governance:** the responsibility of an IT board (COBIT)
 - Architecture governance:** the responsibility of an architecture board.
- Different enterprises relate these governance organisations in different ways.

IT governance organisation

- IT governance responsibility of IT steering committee
- Typical IT governance organisation models are



Architecture governance

- [Governance] of architecture, development and operations to ensure it conforms to pre-defined architectural requirements, principles, policies and models.

Architecture board

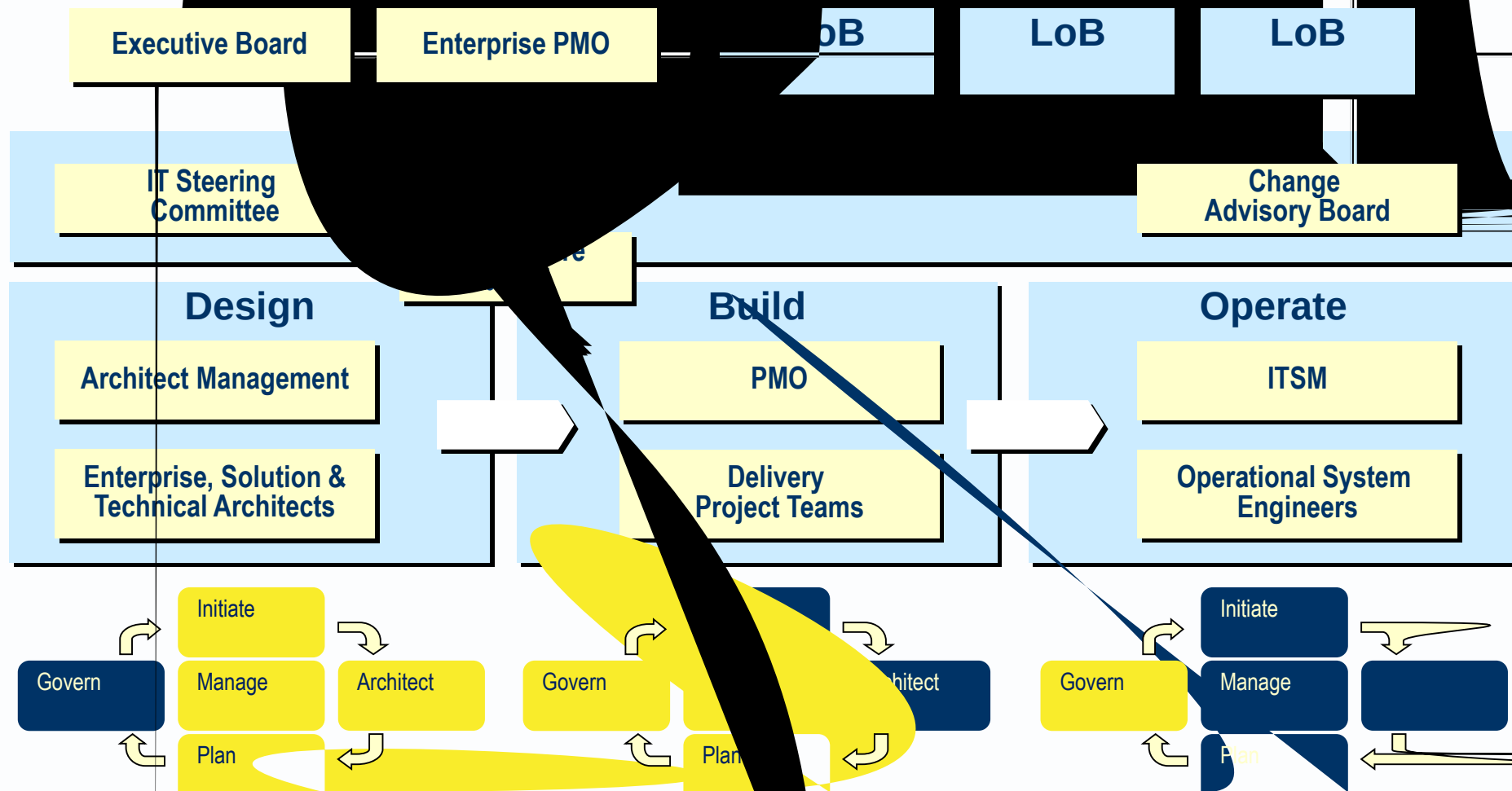
- [An organisation unit] that maintains architecture principles and governance processes, promotes and ensures provision of architecture resources, and reviews compliance reports.

Architecture contract

- [A document] that defines those architectural requirements, principles, policies and specifications that a system should conform to as it is built and when it runs.
- Also defines any architecture stakeholder rights and interests that must be met.

Where do

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- [A work role] the architect who has been nominated by the governance organisation to ensure a system is built and/or run in accord with its architecture contract, to manage risk and to ensure the value of the system to its stakeholders.
- The role may be played by a chief architect or design authority, architecture domain specialist or solution architect, depending on what is to be reviewed.

- **Architecture compliance review**

- [A work process] for monitoring work against architecture aims, directives and documents. Different reviews may be carried out at different points in system design and development.
- Reviews may require a governing architect and/or use an architecture review checklists.

- **Architecture review checklist**

- [An artifact] a standard checklist of questions for an architecture compliance review.
- The questions are general ones, not necessarily mentioned in the architecture contract.

- **Architecture conformance level**

- [A property] showing how well or how much of an architecture contract is met by a system, or an architecture is realised in a system.

- **Architecture compliance level**

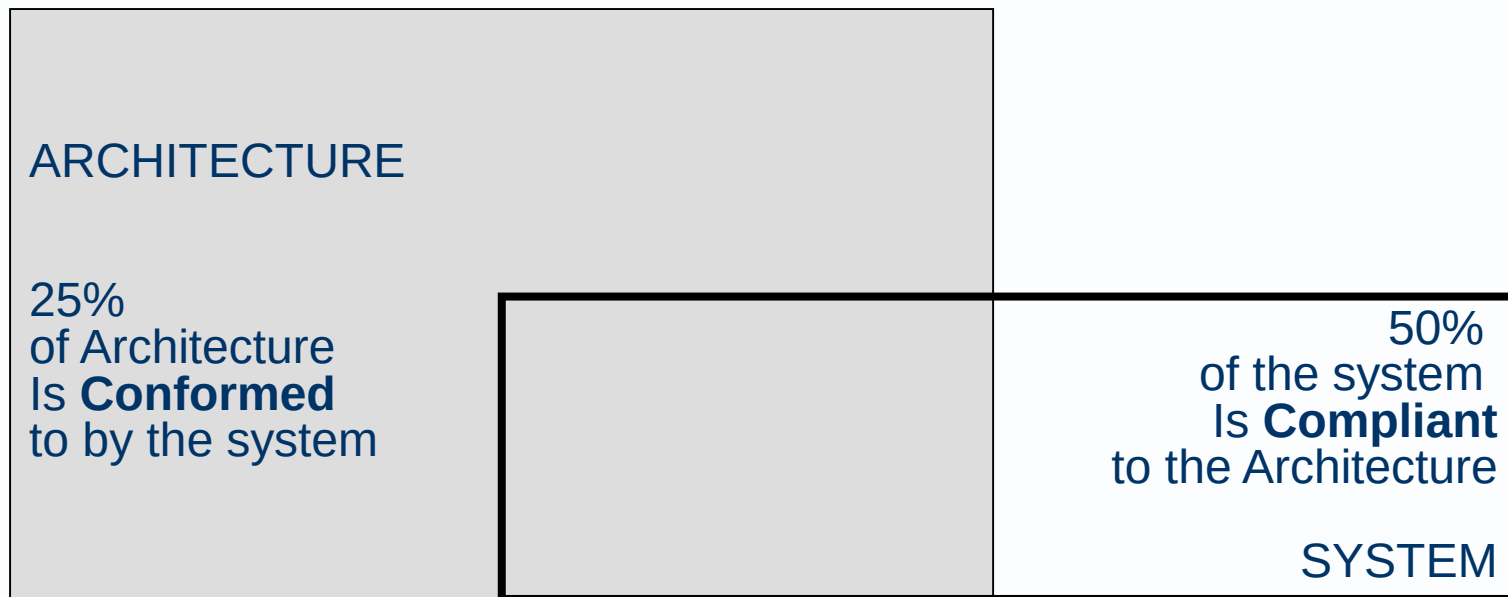
- [A property] showing how well or how much of a system corresponds to its architecture contract and/or description.

- **Dispensation**

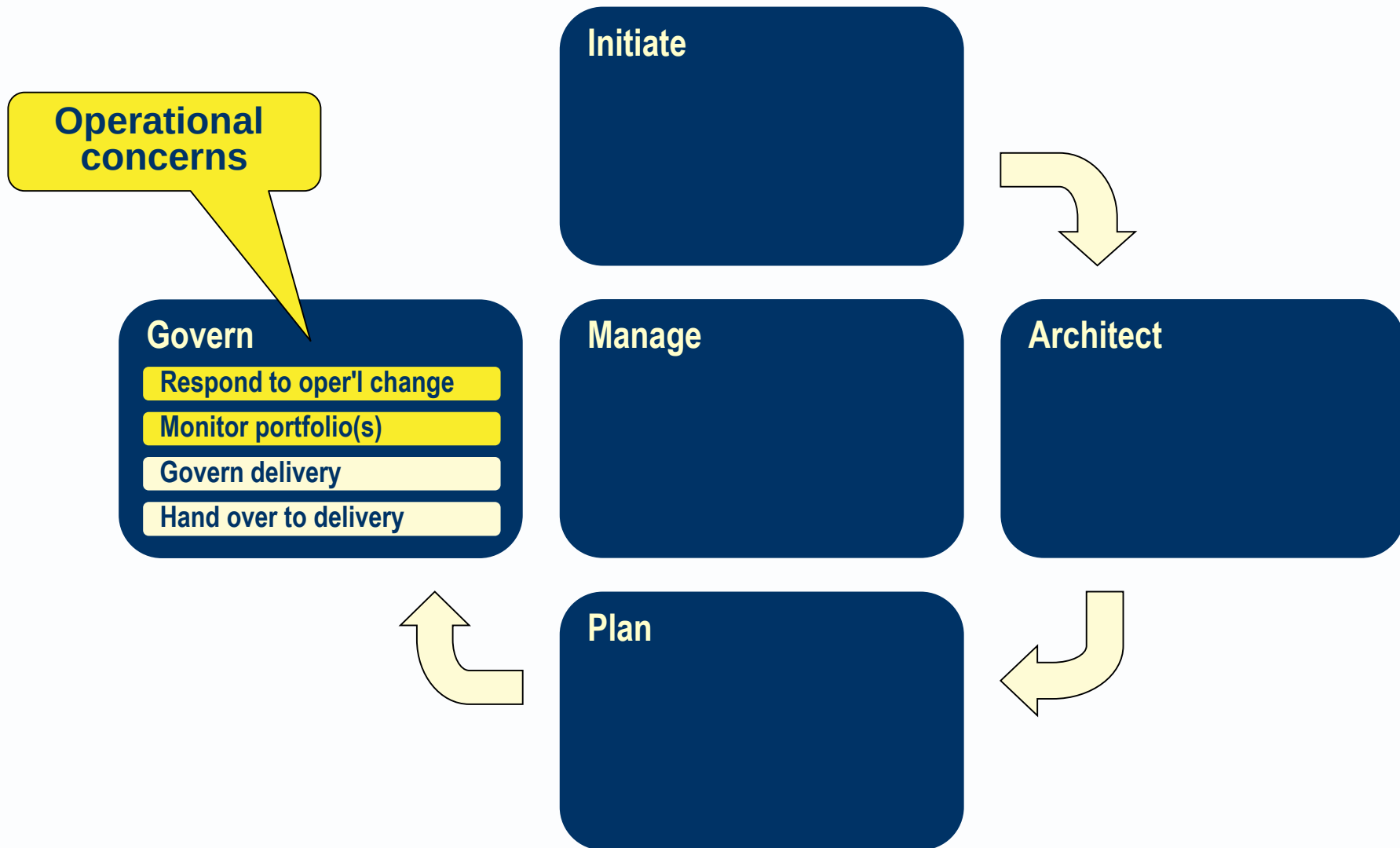
- [A document] a time-bound waiver from the terms of an architecture contract.
- It is granted by a governing architect, and should be reviewed after the specified time.

Conformance v compliance

- Your chance of remembering the difference is 50/50!



Architecture in Operations in the AM process



- The organisation and processes that are needed to manage the architecture description of an operational system.
- Public companies subject to the U.S. Sarbanes-Oxley Act of 2002 are encouraged to adopt COSO and/or COBIT
- Many more appear to have adopted ITIL

	People	Processes	Products
Service Management	Board Service managers Service administrators	System administration Problem/help desk Event monitoring Service level monitoring Exception reporting Exception handling	Service level requirements Service level agreements Service level reports

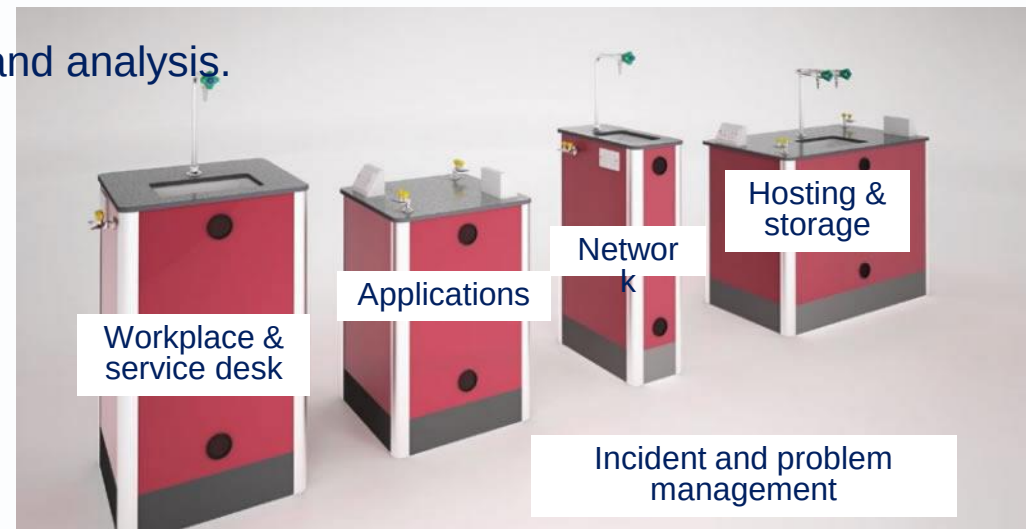
- [A service] provided an IT operations department. E.g.
 - management of user roles and identities,
 - client device configuration,
 - storage administration,
 - network provision, monitoring and analysis,
 - server provision, monitoring and analysis,
 - business activity monitoring,
 - virtualisation,
 - back up & restore,
 - incident and problem management.

Locate Service Offerings by Category

newScale Service Portfolio Library™

 Engineering	 Application Services Service offerings for the application environment, including development, testing, and deployment of new applications. • Application Hosting • Enterprise Application Management	 Consultation Services Service offerings for the consultation environment, including business process reengineering and project consulting. • Business Process Reengineering • Project Consulting
 Workplace Services	 Workplace Services Service offerings for the end user environment, including desktop computing, devices and remote end user experience. • Desktop Computing • Email and Calendaring	 Technical Services (IT to IT) • Mainframe • Network • Servers • Storage

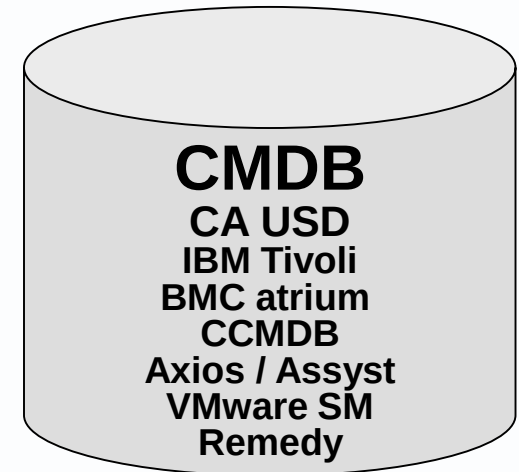
- Workplace and service desk:
Desktop/laptop/mobile configuration and onsite support,
Management of user roles and identities
- Applications:
Support and maintenance
Projects
- Network:
Network and telephony provision, monitoring and analysis
- Hosting and storage:
Data centre, server provision, monitoring and analysis.
Virtualisation, back up & restore
- Incident and problem management



- **ITSM: IT Services Management**
- [A work process] the roles and processes for managing IT infrastructure and the services it provides.
- **IT4IT**
- A product of The Open Group that applies enterprise architecture principles to ITSM.
- It defines a value chain with four primary value streams.
- It decomposes each value stream into functions and defines artifacts that are produced by and exchanged between functions.

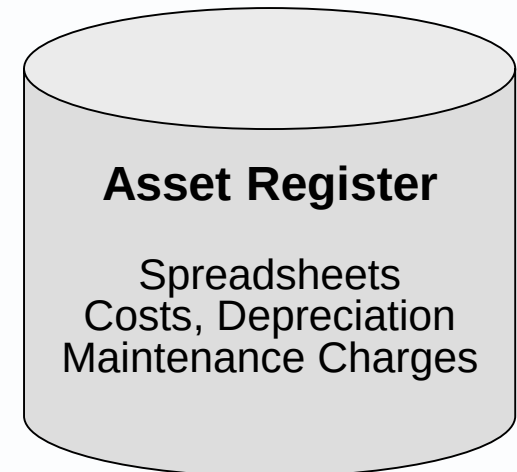
CMDB: Configuration Management Database

- [A data store] a record of configuration item specifications including relationships among configuration items, where the items are significant to ITSM.

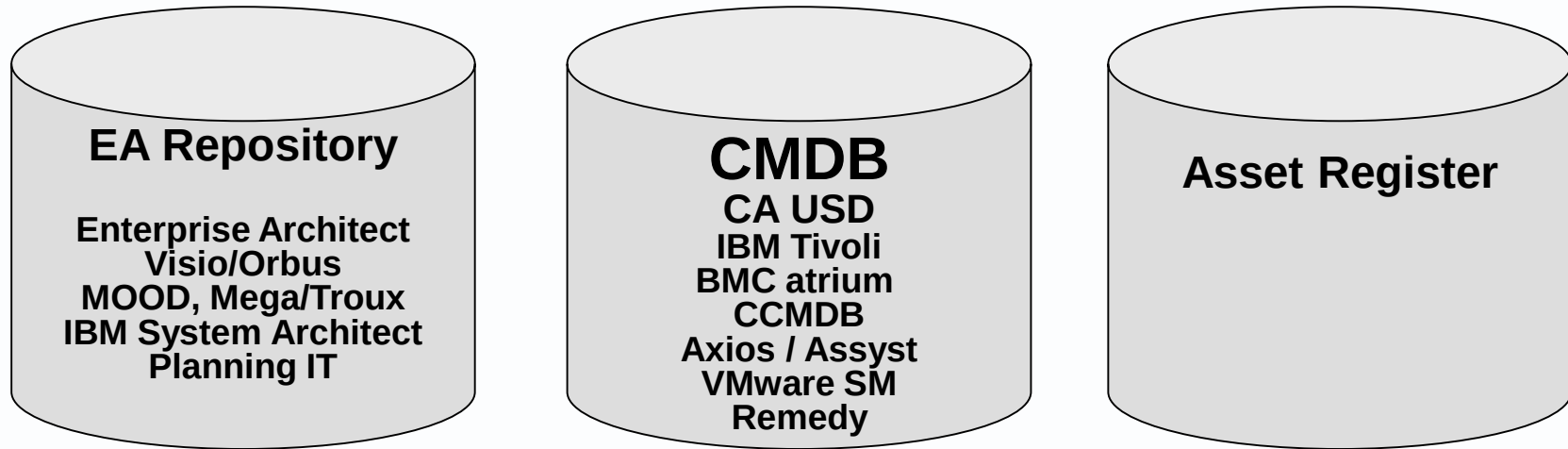


Asset management system

- [A data store] a record of IT assets.
- It is sometimes use to record end user devices, outside of the data centre.
- It may be related to a CMDB.



How many repositories?



- In theory at least, all these configurations are related
- How do they relate? J O of k i Pvpdj > o efp qf fep ebo? J O of k i pl ir of kp ka solution delivery products such as IBM Tivoli CCMDB can provide vital support for this pragmatic approach.
- It facilitates efficient creation and maintenance of the IT architecture by importing auto-af l sbba mmif of k ka ek l il dv m of if p –

Capability Maturity

Capability Maturity

- [A reference model] for evaluating the maturity of an organisation and its processes.
- The first was the capability maturity model (CMM) for software processes.
- The capability maturity model integration (CMMI) widened CMM to cover other processes.
- There are now various maturity models for architecture processes.

- See also the GAO EA management maturity framework.
- <http://www.gao.gov/new.items/d03584g.pdf>

The standard levels in a “maturity model”

Formally

- 5 Optimizing: Focus on process improvement**
- 4 Managed: Process measured and controlled**
- 3 Defined: Process characterized, fairly well understood.**
- 2 Repeatable: Can repeat previously mastered tasks**
- 1 Initial: Unpredictable and poorly controlled**
- 0 Missing: Does not exist**

Processes are

- 5 – continuously improved**
- 4 – measured**
- 3 – documented & followed**
- 2 – repeated**
- 1 – ad hoc (hero-level)**
- 0 - irrelevant**

6 maturity levels

- 1. None
- 2. Initial
- 3. Under development
- 4. Defined
- 5. Managed
- 6. Measured

9 architecture characteristics:

- 1. IT architecture process
- 2. IT architecture development
- 3. Business linkage
- 4. Senior management involvement
- 5. Operating unit participation
- 6. Architecture communication
- 7. IT security
- 8. Architecture Governance
- 9. IT investment and acquisition strategy

2 kinds of maturity rating.

- a weighted mean IT architecture maturity level.
- the percentage achieved at each maturity level for the nine architecture characteristics.